

PARIKSHA365 — STUDY NOTES

Physics — Common Book

SSC, RRB, State PSC, Banks (GA) — General Science

SSC

RRB

BANKS

STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

Examiner's Blueprint — How Physics Gets Tested

⚠ EXAM ALERT

Physics questions in competitive exams fall into a predictable pattern. They are rarely about solving equations — they are almost always about conceptual recall and application. The examiner's favourite moves are: **1. "Why does X happen?" questions** (pressure cooker, ice floating, sky colour) — these are pure understanding questions, and they appear in every paper. **2. "Who discovered / invented X?"** — Röntgen, Faraday, Chadwick, Fleming; these appear 1-2 times per paper. **3. Correction for vision defects** — myopia/hypermetropia and concave/convex lens. Never skipped by examiners. **4. Motor vs Generator** (which is which, which hand rule) — a classic 1-mark gift if you know it. **5. Sound speed and echo distance** — straightforward calculation questions. **6. Alpha, Beta, Gamma radiation properties** — penetration, charge, what stops them. Pure numerics (actual force calculations, wave equation solving) appear rarely. This book builds your conceptual understanding first, then gives you the numbers you must know cold.

Priority order for your study time:

1. Daily-life phenomena (Part X) — "why" questions; one in every paper
 2. Instruments table (Part V) — 1-2 direct questions per paper
 3. Discoverers and inventions (Part W) — constant presence
 4. Sound: speed, echo, ranges (Part D) — easy marks
 5. Light: mirror/lens types, vision defects, Rayleigh scattering (Part D)
 6. Newton's laws with examples (Part A)
 7. Electricity: Ohm's law, Indian supply, AC/DC (Part E)
 8. Electromagnetism: motor/generator/transformer (Part E)
 9. Nuclear: fission/fusion/radiation types (Part F)
 10. Units, constants, formulas (Appendices)
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PART A — MECHANICS

Chapter A1 — SI Units

The scientific world agreed on one system of measurement — the **International System of Units (SI)**. Here are its 7 base units, which all other units are built from.

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

☉ MNEMONIC

****"My Kind Sister Always Keeps Many Candles"*** — Mass (kg), Kind (K = kelvin), Sister (s = second), Always (A = ampere), Keeps (this is a connector), Many (m = metre), Candles (cd = candela). Or more directly: ****metre, kilogram, second, ampere, kelvin, mole, candela**** — "Mr. K Sees Amphibians Killing Many Critters."

Derived Units to Know

Quantity	Unit	Symbol	In base units
Force	newton	N	$\text{kg}\cdot\text{m}/\text{s}^2$
Pressure	pascal	Pa	N/m^2
Energy / Work	joule	J	$\text{N}\cdot\text{m}$
Power	watt	W	J/s
Frequency	hertz	Hz	s^{-1}
Electric charge	coulomb	C	$\text{A}\cdot\text{s}$
Voltage	volt	V	J/C
Resistance	ohm	Ω	V/A

✓ KEY FACT

The commercial unit of electrical energy is 1 kilowatt-hour (kWh) = 1 unit on your electricity bill = 3.6×10^6 joules. 1 horsepower = 746 watts (not 750 — examiners use this to catch those who round up carelessly).

Chapter A2 — Newton's Laws of Motion

Isaac Newton formulated these three laws in 1687 in his *Principia Mathematica*. They governed everything from cricket balls to rockets until Einstein refined them for extreme speeds and masses.

First Law — Law of Inertia

An object at rest stays at rest, and an object moving at constant velocity continues moving at constant velocity, unless acted upon by an external force.

Inertia is the tendency of an object to resist change in its state of motion. Inertia is proportional to mass — a truck is harder to start moving than a bicycle.

⚠ EXAM ALERT

Exam applications of Newton's First Law: seat belts (body wants to continue moving when car stops), pulling a tablecloth quickly (plates stay due to inertia), coins on a card flicked away (coins fall into the glass), dust flying off a beaten carpet. Any "lurching" or "jerking" scenario = First Law.

Second Law — $F = ma$

The rate of change of momentum of an object is proportional to the applied force and in the direction of the force.

Simplified: **Force = mass × acceleration**

- More force → more acceleration.
- More mass → less acceleration for same force.
- SI unit of force: newton (N). One newton accelerates 1 kg at 1 m/s^2 .

Third Law — Action and Reaction

For every action there is an equal and opposite reaction. The two forces act on different objects.

⚠ EXAM ALERT

Third Law applications that appear in exams: **rocket propulsion** (gas pushed backward → rocket pushed forward), **gun recoil** (bullet pushed forward → gun pushes back), **swimming** (you push water backward → water pushes you forward), **helicopter tail rotor** (counters torque from main rotor). Jet engines, walking, and even rowing a boat are Third Law applications.

Equations of Motion (Constant Acceleration)

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$

Where: u = initial velocity, v = final velocity, a = acceleration, t = time, s = displacement.

Chapter A3 — Gravitation

Newton's Universal Law

Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = G \times (m_1 \times m_2) / r^2$$

$G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ (Cavendish first measured this in 1798).

Acceleration Due to Gravity (g)

- g at Earth's surface = 9.8 m/s^2 (use 10 for quick calculations).
- g is **less at the equator** than at the poles — because: (1) the Earth is oblate (bulges at equator, so equatorial surface is farther from the centre), and (2) rotational effect reduces effective gravity.
- g **decreases** as you go higher (away from centre).
- g **decreases** as you go deeper (less mass pulling you).
- **At Earth's centre, $g = 0$.**

✓ KEY FACT

Escape velocity from Earth = **11.2 km/s** (~40,300 km/h). This is the minimum speed an object needs to permanently escape Earth's gravity without further propulsion. Orbital velocity at Earth's surface $\approx 7.9 \text{ km/s}$ — the speed needed to maintain a circular orbit just above the surface.

Kepler's Laws of Planetary Motion

1. **Law of Orbits** — Planets move in elliptical orbits with the Sun at one focus.
2. **Law of Areas** — A line joining a planet to the Sun sweeps out equal areas in equal time intervals (planets move faster when closer to the Sun).
3. **Law of Periods** — $T^2 \propto r^3$ (square of orbital period is proportional to cube of semi-major axis).

Satellites

- A **geostationary satellite** orbits at ~36,000 km above the equator, completing one orbit in exactly 24 hours — so it appears stationary relative to Earth's surface. Used for telecommunications and weather.
- **Weightlessness** in a satellite — the satellite and everything inside are in free fall together. The apparent weight is zero because both the person and the floor accelerate downward at the same rate.

Chapter A4 — Work, Energy, Power

- **Work** = Force $\times$ displacement $\times \cos(\theta)$. Work is done only when displacement occurs in the direction of force. Unit: joule (J).
- **Kinetic Energy (KE)** = $\frac{1}{2}mv^2$
- **Potential Energy (gravitational PE)** = mgh
- **Power** = Work done / Time taken. Unit: watt (W). $P = F \times v$.

Conservation of Energy

Energy cannot be created or destroyed. It transforms from one form to another. Total energy of an isolated system is constant.

Classic example: A pendulum at its highest point has maximum PE and zero KE. At the lowest point, PE converts entirely to KE. The total (PE + KE) stays constant throughout the swing (ignoring air resistance).

PART B — PROPERTIES OF MATTER

Chapter B1 — Pressure, Fluids, Buoyancy

- **Pressure** = Force / Area. SI unit: pascal (Pa). A sharp knife cuts better than a blunt one because the same force is applied over a smaller area.
- **Atmospheric pressure** at sea level = 101,325 Pa $\approx$ 1 atm = 760 mm Hg.
- Atmospheric pressure decreases with altitude (less air column above).

Pascal's Law

Pressure applied to an enclosed fluid transmits equally in all directions. This is why hydraulic systems (car brakes, JCB excavators, dental chairs) can multiply force enormously with a small input.

Archimedes' Principle

The upward buoyant force on a body immersed in fluid equals the weight of fluid displaced by the body.

A steel ship floats because it is hollow — its effective density (steel + air) is less than water, so it displaces enough water for the buoyant force to equal its weight. A solid steel ball sinks because its density is much greater than water.

⚠ EXAM ALERT

Ice floats on water — this is anomalous expansion. Water expands on freezing (unusual behaviour). Ice density is 0.92 g/cm³ vs water at 1.0 g/cm³. This is why fish survive winter — ice floats and forms an insulating layer; liquid water at 4 °C stays below, where fish live.

Chapter B2 — Surface Tension, Viscosity, Capillarity

- **Surface tension** — why water forms droplets (spherical = minimum surface area for a given volume), why insects can walk on water, why a needle can float on still water. Detergent reduces surface tension — the soap molecule disrupts the surface layer.
 - **Viscosity** — internal friction in fluids. Honey > water > petrol. Viscosity decreases with temperature (why engine oil thins when hot).
 - **Capillarity** — rise or fall of liquid in narrow tubes due to surface tension vs gravity. Water rises in a glass capillary (adhesion to glass > cohesion between water molecules). Mercury falls in a glass capillary. Plants use capillary action to draw water up roots and stems.
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PART C — HEAT AND THERMODYNAMICS

Chapter C1 — Temperature Scales

Scale	Freezing of water	Boiling of water
Celsius (°C)	0	100
Fahrenheit (°F)	32	212
Kelvin (K)	273.15	373.15

Conversion: $^{\circ}\text{F} = (9/5 \times ^{\circ}\text{C}) + 32$. Kelvin = Celsius + 273.15.

Absolute zero (0 K = -273.15°C) — theoretically the coldest possible temperature; particles have minimum possible energy.

Chapter C2 — Heat Transfer

Mode	Medium needed	Mechanism	Examples
Conduction	Solid (mainly)	Particle vibration, electron flow	Metal rod heating, iron; ice cube melting in hand
Convection	Fluid (liquid/gas)	Bulk movement of fluid	Boiling water, sea breeze, room air conditioning
Radiation	None — vacuum OK	Electromagnetic waves	Sun warming Earth, warmth from a fire

✓ KEY FACT

The Sun's heat reaches Earth through the vacuum of space — this is radiation (electromagnetic waves). Only radiation requires no medium. This is why a thermos (vacuum flask) has a silvered interior — it prevents heat loss by all three mechanisms: vacuum stops conduction and convection, silvering reflects radiation.

Chapter C3 — Specific Heat and Latent Heat

- **Specific heat** — heat energy needed to raise 1 kg of a substance by 1°C (or 1 K).
- Water: $4184 \text{ J/kg}\cdot^{\circ}\text{C}$ — the highest among common substances. This is why coastal climates are milder — oceans absorb and release enormous heat with little temperature change.
- Iron: $\sim 449 \text{ J/kg}\cdot^{\circ}\text{C}$. Aluminium: $\sim 900 \text{ J/kg}\cdot^{\circ}\text{C}$.

- **Latent heat** — heat absorbed or released during a phase change (state change) WITHOUT any temperature change.
- Latent heat of fusion of ice: 334 J/g (ice → water at 0 °C).
- Latent heat of vaporisation of water: 2260 J/g (water → steam at 100 °C).

⚠ EXAM ALERT

Steam burns are worse than boiling water burns at the same 100 °C — because steam releases its latent heat of vaporisation (2260 J/g) when it condenses on your skin, in addition to the heat from cooling. We sweat to cool down because sweat evaporating absorbs latent heat of vaporisation from the skin surface.

Chapter C4 — Laws of Thermodynamics

1. **Zeroth Law** — If system A is in thermal equilibrium with system B, and B with C, then A and C are also in equilibrium. This defines temperature.
2. **First Law** — $\Delta U = Q - W$. Energy is conserved. Heat added to a system equals change in internal energy plus work done by the system.
3. **Second Law** — Heat cannot spontaneously flow from cold to hot. Entropy (disorder) of an isolated system always increases.
4. **Third Law** — At absolute zero, entropy of a perfect crystal is zero. Absolute zero is unreachable in practice.

A refrigerator moves heat from cold to hot — but only by consuming electricity (external work). This is the Second Law in action.

PART D — WAVES, SOUND, AND LIGHT

Chapter D1 — Sound

Sound is a mechanical wave — it requires a medium to travel. It cannot travel through vacuum. That is why space is silent.

Speed of Sound

Medium	Speed (approximate)
Air at 0 °C	331 m/s
Air at 20 °C	343 m/s
Water	~1500 m/s
Steel	~5000 m/s

Sound travels faster in solids than in liquids, and faster in liquids than in gases. Speed increases with temperature.

Frequency Ranges

Range	Frequency	Who uses it
Infrasonic	Below 20 Hz	Whales, elephants, earthquakes
Audible (human)	20 Hz to 20,000 Hz	Humans
Ultrasonic	Above 20,000 Hz	Bats, dolphins, dogs; medical sonography, SONAR

✓ KEY FACT

Echo needs a minimum distance of ****17 metres**** between you and the reflecting surface. Why? Sound takes 0.1 seconds to reach the reflector and return (minimum time for the human ear to distinguish echo from original). At 343 m/s, that is $343 \times 0.1 / 2 = 17.15$ m.

Doppler Effect

The apparent frequency of a sound changes when the source and observer are in relative motion. When the source approaches, frequency appears higher (pitch rises). When the source moves away, frequency appears lower (pitch falls). Classic example: an ambulance siren changes pitch as it passes you.

SONAR

Sound Navigation and Ranging — uses ultrasonic pulses to detect underwater objects and map the ocean floor. The time for the echo to return, multiplied by the speed of sound in water, gives twice the distance to the object.

Chapter D2 — Light and Optics

Light is an electromagnetic wave that travels at $c = 3 \times 10^8 \text{ m/s}$ in vacuum. This is the fastest speed in the universe.

Speed of light in other media: water $\sim 2.25 \times 10^8 \text{ m/s}$, glass $\sim 2 \times 10^8 \text{ m/s}$. Light slows down in denser media.

Refractive index (n) = speed of light in vacuum / speed of light in medium. For glass ≈ 1.5 , water ≈ 1.33 , diamond ≈ 2.42 .

Reflection

- Angle of incidence = angle of reflection.
- Incident ray, reflected ray, and normal are in the same plane.

Refraction (Snell's Law)

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Light bends toward the normal when entering a denser medium and away from the normal when entering a less-dense medium.

Total Internal Reflection (TIR) — when light travels from a denser to a less-dense medium at an angle greater than the critical angle, it is completely reflected back. This is what gives diamonds their sparkle (very small critical angle $\sim 24^\circ$) and what optical fibres rely on.

Mirrors — Which Type Does What

Mirror	Image formed	Used in
Plane	Same size, virtual, erect, laterally inverted	Looking glass
Concave (converging)	Depends on object position; real+inverted if beyond focus	Shaving mirror, headlights, dentist's mirror, reflecting telescopes
Convex (diverging)	Always virtual, erect, diminished	Rear-view mirrors in vehicles, security mirrors

⚠ EXAM ALERT

Rear-view mirrors in vehicles are always convex — they give a wider field of view (the image is diminished but you see more). The "objects in mirror are closer than they appear" warning is because the convex mirror makes distant objects appear farther away. A driver checking a concave rear-view mirror would get a misleading sense of distances.

Lenses — Which Type Does What

Lens	Image type	Used for
Convex (converging)	Real or virtual depending on position	Magnifying glass, cameras, projectors, microscopes, eye's own lens
Concave (diverging)	Always virtual, erect, diminished	Correcting myopia (short-sightedness)

Lens formula: $1/v - 1/u = 1/f$. Power of lens: $P = 1/f$ (metres). Unit: dioptre (D).

Defects of Vision and Corrections

Defect	What happens	Correction
Myopia (short-sightedness)	Image forms in front of retina; can see near but not far	Concave (diverging) lens
Hypermetropia (long-sightedness)	Image forms behind retina; can see far but not near	Convex (converging) lens
Presbyopia	Eye lens stiffens with age; cannot adjust	Bifocal lens
Astigmatism	Cornea unevenly curved	Cylindrical lens
Cataract	Lens becomes opaque	Surgery to replace lens

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****Myopia → Concave, Hypermetropia → Convex****. Memory trick: the words ****Myopia**** and ****Concave**** both have fewer common letters to remember — just link the ****M**yopic** person needs a ****C**oncave** lens. Or: Myopia = "My can't see far" → need to diverge the light more → concave.

Dispersion and Colour

- **Dispersion** — white light splits into its spectrum (VIBGYOR) when passing through a prism. Newton first demonstrated this in 1666.
- **VIBGYOR**: Violet, Indigo, Blue, Green, Yellow, Orange, Red — in order of decreasing wavelength (increasing frequency). Violet has highest frequency in visible range.

- **Rainbow** — caused by refraction + total internal reflection + dispersion inside raindrops.
- **Blue sky** — Rayleigh scattering. Shorter (blue) wavelengths scatter $\sim 5\times$ more than longer (red) wavelengths. So when sunlight enters the atmosphere, blue light scatters in all directions, making the whole sky look blue.
- **Red sunset** — at sunrise/sunset, sunlight travels through much more atmosphere to reach your eyes. Blue light has already scattered away; red, which scatters least, dominates.

✓ KEY FACT

Diamond sparkles because its refractive index is very high (2.42), giving a very small critical angle ($\sim 24^\circ$). Most light rays entering the diamond hit a face at more than 24° and undergo total internal reflection repeatedly — they bounce inside the diamond and finally exit from the top. This internal bouncing creates the famous sparkle.

Electromagnetic Spectrum

From lowest to highest frequency (highest to lowest wavelength):

Radio → Microwaves → Infrared → Visible light → Ultraviolet → X-rays → Gamma rays

Band	Wavelength	Key use
Radio	km to m	AM/FM radio, TV broadcasting
Microwave	cm to mm	Microwave ovens, mobile phones, radar
Infrared (IR)	μm	TV remotes, thermal cameras, heat sensing
Visible	400–700 nm	Human vision (VIBGYOR)
Ultraviolet (UV)	$<400\text{ nm}$	Sterilisation, vitamin D synthesis, fluorescence
X-rays	0.01–10 nm	Medical imaging, airport security
Gamma	$<0.01\text{ nm}$	Cancer radiotherapy, sterilising medical equipment

PART E — ELECTRICITY AND MAGNETISM

Chapter E1 — Current Electricity

- **Charge (q)** — unit: coulomb (C).
- **Current (I)** = charge per second = q/t . Unit: ampere (A).
- **Voltage / Potential difference (V)** — the "electrical pressure" that drives current. Unit: volt (V).
- **Resistance (R)** — opposition to current. Unit: ohm (Ω).

Ohm's Law

$$V = I \times R$$

At constant temperature, current through a conductor is proportional to the voltage across it.

Power consumed: $P = V \times I = I^2 R = V^2/R$. Unit: watt (W).

Household Electricity in India

230 V, 50 Hz, AC. The 230 V is the RMS voltage. USA uses 120 V, 60 Hz.

- **AC (Alternating Current)** — current reverses direction periodically (50 times per second in India = 50 Hz). Used for transmission over long distances because voltage can be stepped up or down using transformers.
- **DC (Direct Current)** — current flows in one direction only. Used in batteries, electronic devices.

⚠ EXAM ALERT

Why is AC preferred for power transmission? Transformers can step up voltage for long-distance transmission (reducing current $\rightarrow$ reducing heat loss in wires: $P = I^2 R$). At the destination, step it back down for safe home use. Transformers work only on AC, not DC — because DC doesn't create a changing magnetic flux, so no induction occurs.

Circuits and Safety

- **Fuse** — a thin wire of low-melting-point alloy that melts when current exceeds a safe limit, breaking the circuit.
 - **MCB (Miniature Circuit Breaker)** — modern alternative; trips a switch instead of burning a wire; can be reset.
 - **Short circuit** — accidental path of near-zero resistance; causes very high current $\rightarrow$ heat $\rightarrow$ fire.
 - **Earthing** — safety connection to the ground. Any leakage current flows harmlessly into the ground instead of through a person.
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Chapter E2 — Magnetism and Electromagnetic Induction

Magnets

- A bar magnet has two poles — North and South. Like poles repel; unlike poles attract.
- Earth is a giant magnet. The geographic North Pole of Earth is near the magnetic South Pole (that's why the North-seeking pole of a compass needle points north — it's attracted to Earth's magnetic south).
- Magnetic field lines run from North to South outside the magnet (and South to North inside).

Electromagnetic Induction — Faraday's Discovery (1831)

Moving a magnet near a coil of wire induces a voltage (EMF) in the coil, and therefore a current if the circuit is complete.

✓ KEY FACT

This is the foundation of everything electrical we use. Generators in power stations, microphones, guitar pickups, wireless charging — all are applications of electromagnetic induction. Michael Faraday made this discovery in 1831, and it changed civilisation.

Lenz's Law

The direction of the induced current is always such that it opposes the change (motion or flux change) causing it. Think of it as electromagnetic braking — nature resists change.

Motor and Generator

Device	Converts	Principle	Hand Rule
Electric motor	Electrical → Mechanical	Current-carrying conductor in a magnetic field experiences a force	Fleming's Left-Hand Rule
Generator / Dynamo	Mechanical → Electrical	Moving conductor in a magnetic field induces current	Fleming's Right-Hand Rule

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****Fleming's Left-Hand Rule for Motor****: Hold your left hand with thumb, index finger, and middle finger all at right angles to each other. Index finger = direction of magnetic Field, Middle finger = direction of Current, Thumb = direction of Thrust (motion/force on conductor). For a generator, use the Right hand with the same logic — Motion thumb, Field index, Current middle. Motor = Left = I drive the ****L****eft side. Generator = Right.

Transformer

Changes the voltage of AC supply.

- **Step-up transformer:** increases voltage, decreases current.
- **Step-down transformer:** decreases voltage, increases current.

Works on mutual induction — the changing current in the primary coil creates a changing magnetic field that induces a current in the secondary coil. Transformers work only with AC.

PART F — MODERN PHYSICS

Chapter F1 — Atomic Models

Model	Year	Proposer	Key idea
Plum - pudding	1897	J.J. Thomson	Atom = positive sphere with electrons embedded like plums
Nuclear	1911	Ernest Rutherford	Most of atom is empty space; tiny dense nucleus at centre
Bohr model	1913	Niels Bohr	Electrons orbit in fixed quantised energy levels
Quantum mechanical	1926	Schrödinger / Heisenberg	Electrons exist as probability clouds (orbitals)

Rutherford used the famous gold-foil (alpha-scattering) experiment — firing alpha particles at gold foil. Most passed through (empty space), some deflected slightly (electrons), and a few bounced straight back (nucleus). He concluded the atom is mostly empty with a tiny, dense, positive nucleus.

Chapter F2 — Subatomic Particles

Particle	Charge	Relative mass	Discoverer	Year
Electron	-1	1/1837	J.J. Thomson	1897
Proton	+1	1	Rutherford	1919
Neutron	0	~1 (slightly > proton)	James Chadwick	1932

✓ KEY FACT

The heaviest sub-atomic particle is the **neutron** (just slightly heavier than the proton). Three quick-recall facts: electron → Thomson 1897, proton → Rutherford 1919, neutron → Chadwick 1932. The first Nobel Prize in Physics (1901) went to Röntgen for X-rays.

Chapter F3 — Radioactivity

Henri Becquerel discovered radioactivity in 1896 (from uranium). Marie and Pierre Curie then isolated polonium (named after Marie's homeland Poland) and radium in 1898. Marie Curie won two Nobel Prizes — Physics 1903 and Chemistry 1911.

Three Types of Radiation

Type	Symbol	Nature	Charge	Penetrating power	Stopped by
Alpha	α	Helium nucleus (2p + 2n)	+2	Lowest	Paper, few cm of air
Beta	β	Fast electron	-1	Medium	3mm aluminium
Gamma	γ	High-energy EM wave (photon)	0	Highest	Several cm of lead

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Penetrating power order: $\alpha < \beta < \gamma$ (alphabetical = weakest to strongest). What stops them: **Paper, Aluminium, Lead** — think PAL. Alpha needs a PAL to stop it (paper stops alpha). Aluminium stops beta. Lead stops gamma.

Chapter F4 — Nuclear Reactions

Fission

A heavy nucleus (Uranium-235 or Plutonium-239) splits into smaller fragments + 2-3 neutrons + enormous energy.

Each released neutron can trigger more fissions → chain reaction. In a nuclear reactor, control rods (cadmium or boron) absorb excess neutrons to control the rate. In an atomic bomb, the chain reaction is uncontrolled.

Fusion

Light nuclei (hydrogen isotopes — deuterium and tritium) combine to form helium + energy.

This is the source of the Sun's energy. Fusion releases far more energy per unit mass than fission, but controlled fusion on Earth has not yet been achieved commercially. The ITER project in France aims to demonstrate controlled fusion.

⚠ EXAM ALERT

"Sun's energy comes from nuclear fusion" — this appears in almost every exam. Hydrogen (specifically deuterium and tritium) fuses to form helium, releasing energy according to Einstein's $E = mc^2$. The mass of the helium product is slightly less than the combined mass of the inputs — that tiny mass difference becomes an enormous amount of energy.

E = mc²

Einstein's mass-energy equivalence. Even a tiny mass (m) holds an enormous amount of energy (c = 3 × 10⁸ m/s, so c² = 9 × 10¹⁶). Burning 1 gram of matter completely would yield 9 × 10¹³ joules — equivalent to 20,000 tons of TNT.

Radiocarbon Dating

Carbon-14 (¹⁴C) is a radioactive isotope of carbon with half-life of **5730 years**. Living organisms maintain a constant ¹⁴C/¹²C ratio by continuously taking in carbon. When they die, ¹⁴C decays without being replenished. Measuring the remaining ¹⁴C ratio tells us the age of the organism. Used in archaeology (up to ~50,000 years back).

Chapter F5 — Semiconductors

Type	Resistivity	Band gap	Examples
Conductor	Very low	None	Copper, silver, aluminium
Semiconductor	Intermediate	Small (~1 eV)	Silicon (Si), Germanium (Ge)
Insulator	Very high	Large	Glass, rubber, plastic

- **N-type semiconductor** — doped with pentavalent atoms (phosphorus, arsenic); extra electrons are charge carriers.
- **P-type semiconductor** — doped with trivalent atoms (boron, aluminium); "holes" are charge carriers.
- **Diode** (P-N junction) — allows current in one direction only.
- **LED (Light Emitting Diode)** — emits light when forward current flows.
- **Transistor** — invented 1947 by Bardeen, Brattain, Shockley (Bell Labs); Nobel 1956. Functions as amplifier and switch. Foundation of all modern electronics.
- **Integrated Circuit (IC)** — many transistors on one silicon chip. Invented by Jack Kilby, 1958; Nobel 2000.

PART V — INSTRUMENTS AND WHAT THEY MEASURE

Instrument	Measures
Ammeter	Electric current (connected in series)
Voltmeter	Potential difference / voltage (connected in parallel)
Galvanometer	Very small electric currents
Multimeter	Voltage, current, and resistance
Wattmeter	Electric power
Energy meter	Electrical energy consumed (kWh)
Ohmmeter	Electrical resistance
Thermometer	Temperature
Pyrometer	Very high temperatures (furnaces, molten metal)
Calorimeter	Heat exchanged in reactions
Barometer	Atmospheric pressure
Manometer	Gas pressure
Hygrometer / Psychrometer	Relative humidity
Anemometer	Wind speed
Rain gauge (Pluviometer)	Rainfall
Seismograph	Earthquake intensity and location
Lactometer	Purity / density of milk
Hydrometer	Density of liquids
Saccharometer	Sugar content of a solution
Salinometer	Salt concentration in water
Speedometer	Speed of a vehicle
Odometer	Distance travelled by a vehicle
Tachometer	Rotational speed (RPM) of a shaft or engine
Sextant	Angular altitude of celestial bodies (navigation)
Theodolite	Horizontal and vertical angles (land surveying)

Instrument	Measures
Audiometer	Hearing sensitivity (threshold of hearing)
Stethoscope	Heart and lung sounds
Sphygmomanometer	Blood pressure
Cardiograph (ECG)	Electrical activity of the heart
Electroencephalograph (EEG)	Electrical activity of the brain
Tonometer	Eye pressure (glaucoma screening)
Endoscope	Images from inside the body
Periscope	Viewing over obstacles (used in submarines)
Spectrometer	Wavelengths of light / analysis of light spectrum
Refractometer	Refractive index of a substance
Photometer	Intensity of light
Bolometer	Power of incident electromagnetic radiation
Oscilloscope	Waveform of electrical signals

PART W — DISCOVERIES AND INVENTIONS IN PHYSICS

Discovery / Invention	Scientist	Year	Country
Compound microscope	Zacharias Janssen	1590	Netherlands
Telescope (refractor)	Galileo Galilei	1609	Italy
Reflecting telescope	Isaac Newton	1668	England
Laws of motion, universal gravitation	Isaac Newton	1687	England
Electrostatic machine (Wimshurst)	Robert Van de Graaff	1929	USA
Barometer	Evangelista Torricelli	1643	Italy
Steam engine (improved)	James Watt	1769	Scotland
Battery (voltaic pile)	Alessandro Volta	1800	Italy
Electromagnetic induction	Michael Faraday	1831	England
Laws of thermodynamics (conservation of energy)	Helmholtz, Joule, Mayer	1840s	Germany, England
Telephone	Alexander Graham Bell	1876	Scotland/USA
Electric light bulb (incandescent)	Thomas Edison	1879	USA
Wireless (radio) transmission	Guglielmo Marconi	1895	Italy
Wireless waves (millimetre waves)	Jagadish Chandra Bose	1895	India
X-rays	Wilhelm Röntgen	1895	Germany
Radioactivity	Henri Becquerel	1896	France
Polonium + Radium	Marie and Pierre Curie	1898	Poland/France
Quantum hypothesis	Max Planck	1900	Germany
Aeroplane (first flight)	Wright Brothers	1903	USA
Special Theory of Relativity, photoelectric effect	Albert Einstein	1905	Germany
Bohr atomic model	Niels Bohr	1913	Denmark
Television (mechanical)	John Logie Baird	1925	Scotland
Raman Effect	C.V. Raman	1928	India

Discovery / Invention	Scientist	Year	Country
Cyclotron (particle accelerator)	Ernest Lawrence	1929	USA
Nuclear chain reaction	Enrico Fermi	1942	Italy/USA
Transistor	Bardeen, Brattain, Shockley	1947	USA
Holography	Dennis Gabor	1948	Hungary/UK
Laser	Theodore Maiman	1960	USA
Integrated circuit	Jack Kilby	1958	USA
Optical fibre communication	Charles Kao	1966	China/UK
Gravitational waves detected (LIGO)	LIGO team (Thorne, Weiss, Barish)	2015/2017	USA
Higgs boson confirmed	CERN experiments	2012	International
Bose - Einstein statistics	S.N. Bose + Einstein	1924	India/Germany
Chandrasekhar limit	S. Chandrasekhar	1930	India/USA

⚠ EXAM ALERT

Indian scientists to remember for physics: **C.V. Raman** (Raman Effect, Nobel Prize in Physics 1930 — the only Nobel in Physics won by an Indian as of 2026), **Jagadish Chandra Bose** (millimetre/microwave waves, 1895 — before Marconi in some interpretations), **Satyendra Nath Bose** (Bose-Einstein statistics, 1924 — the "boson" in Higgs boson is named after him), **Subrahmanyan Chandrasekhar** (Chandrasekhar limit, Nobel 1983).

PART X — DAILY-LIFE PHYSICS ("WHY" QUESTIONS)

Examiners love this section. One to two questions on everyday physics phenomena appear in almost every competitive exam. Learn the explanation for each.

Phenomenon	Explanation
Pressure cooker cooks faster	Higher pressure raises the boiling point of water to $\sim 120^\circ\text{C}$; food cooks faster at higher temperature
Cooking takes longer on mountains	Atmospheric pressure is lower at high altitude; water boils at lower temperature ($\sim 70^\circ\text{C}$ on Everest); food cooks slower
Ice floats on water	Anomalous expansion — water expands on freezing; ice density (0.92 g/cm^3) $<$ water density (1.0 g/cm^3)
Fish survive under frozen lake	Ice forms on top and insulates the water below; liquid water at $\sim 4^\circ\text{C}$ stays at the bottom where fish live
Mirage in desert	Light bends due to temperature gradient in hot air near the ground — total internal reflection of sky light
Stars twinkle, planets don't	Stars are point sources — atmospheric turbulence causes their single ray to flicker. Planets are extended discs — their many rays average out, no twinkling
Sky is blue	Rayleigh scattering — shorter (blue) wavelengths scatter much more than longer (red) wavelengths
Sunset and sunrise are red/orange	Sunlight travels a long path through the atmosphere; blue has already scattered away; red, which scatters least, survives to reach your eyes
Rainbow forms after rain	Water droplets act as tiny prisms — refraction + total internal reflection + dispersion separate white light into VIBGYOR
Diamond sparkles	Very high refractive index (2.42) $\rightarrow$ small critical angle ($\sim 24^\circ$) $\rightarrow$ total internal reflection of most rays $\rightarrow$ sparkle
Soap film shows colours	Thin-film interference — light reflects from both surfaces of the thin film; they interfere constructively/destructively at different wavelengths
Why we sweat to cool	Evaporation of sweat absorbs latent heat of vaporisation from the skin surface
Steam burn worse than water burn	Steam at 100°C releases latent heat of vaporisation (2260 J/g) when it condenses on skin, plus heat from cooling
We slip on ice	Pressure of body weight on ice slightly lowers its melting point; a thin film of water forms $\rightarrow$ low friction

Phenomenon	Explanation
Spoon appears bent in water	Refraction — light bends when crossing from water to air; the image appears to shift
Lightning seen before thunder	Light travels at 3×10^8 m/s, sound at ~ 340 m/s; light arrives almost instantly, sound takes ~ 3 seconds per km
Ceiling fan cools us	Increases evaporation of sweat from skin; does not actually cool the room air
Hot tea in saucer cools faster	Larger surface area increases rate of evaporation and radiation
Pendulum clock loses time in summer	Pendulum rod expands with heat $\rightarrow$ longer pendulum $\rightarrow$ longer period $\rightarrow$ clock runs slow
Birds sit on high-tension wires safely	Both feet are on the same wire at the same potential; no potential difference between them $\rightarrow$ no current through bird
A 3-pin plug is safer	Earth pin diverts any leakage current to the ground instead of through the user
Microwave heats food	Microwaves make water molecules vibrate (dielectric heating); heating from inside out
Helicopter needs a tail rotor	Main rotor rotation creates torque on the body (Newton's 3rd Law); tail rotor counters this torque
Geostationary satellite stays above same spot	Orbital period is 24 hours, same as Earth's rotation; appears stationary from Earth
Satellites feel weightless	Satellite and astronaut are in free fall together; apparent weight = 0 (no contact force needed)
Echo minimum distance is 17 m	Human ear needs 0.1 s gap to distinguish echo from original sound; at 340 m/s, minimum round trip = 34 m $\rightarrow$ 17 m one way
Mountaineers carry oxygen cylinders	Atmospheric pressure at high altitude is very low; partial pressure of oxygen is insufficient for breathing

PART AA — KEY CONSTANTS AND FORMULAS

Physical Constants

Constant	Symbol	Value
Speed of light in vacuum	c	$3 \times 10^8 \text{ m/s}$
Acceleration due to gravity (Earth surface)	g	9.8 m/s^2
Universal gravitational constant	G	$6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
Planck's constant	h	$6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
Charge of electron	e	$1.6 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.11 \times 10^{-31} \text{ kg}$
Mass of proton	m_p	$1.67 \times 10^{-27} \text{ kg}$
Avogadro's number	N_a	6.022×10^{23}
Gas constant	R	8.314 J/mol/K
Boltzmann constant	k	$1.38 \times 10^{-23} \text{ J/K}$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$
Atmospheric pressure	—	$101,325 \text{ Pa} \approx 760 \text{ mm Hg}$

Key Non-SI Conversions

- 1 horsepower (HP) = 746 watts
- 1 kWh (1 unit of electricity) = 3.6×10^6 joules
- 1 calorie = 4.184 joules
- 1 eV (electron volt) = $1.6 \times 10^{-19} \text{ J}$
- 1 light-year = 9.46×10^{15} metres
- 1 ångström = 10^{-10} metres
- $0^\circ\text{C} = 273.15 \text{ K} = 32^\circ\text{F}$

Important Numerical Values to Memorise

- Escape velocity from Earth = **11.2 km/s**
- Orbital velocity at Earth surface = **7.9 km/s**
- Geostationary orbit altitude = **36,000 km**
- Speed of sound in air (20°C) = **343 m/s**
- Specific heat of water = **4184 J/kg/°C**
- Latent heat of fusion of ice = **334 J/g**

- Latent heat of vaporisation of water = **2260 J/g**
- India household supply = **230 V, 50 Hz AC**
- Refractive index of diamond = **2.42**
- Refractive index of water = **1.33**

Formula Quick Reference

Mechanics: $F = ma$ | $W = mgh$ | $KE = \frac{1}{2}mv^2$ | $P = W/t$ | $v^2 = u^2 + 2as$

Heat: $Q = mc\Delta T$ | $Q = mL$ | $F = (9/5)C + 32$ | $K = C + 273$

Light: $n = c/v_{\text{medium}}$ | $\sin(\theta_c) = 1/n$ | $1/v - 1/u = 1/f$ | $P = 1/f$ (diopetre)

Electricity: $V = IR$ | $P = VI = I^2R$ | $E = Pt$ | series: $R = R_1 + R_2$ | parallel: $1/R = 1/R_1 + 1/R_2$

Modern: $E = mc^2$ | $E = hf$ | $\lambda = h/p$ (de Broglie)

APPENDIX A — 25-Question Mini-Mock

Attempt all 25 without looking at answers. Time yourself — you should finish in 20 minutes.

1. What is the SI unit of pressure?
2. Who discovered electromagnetic induction and in what year?
3. Escape velocity from Earth is approximately how much?
4. Why does a pressure cooker cook food faster than an open pot?
5. Which type of lens corrects myopia?
6. Speed of sound in air at 0 °C?
7. Which type of mirror is used as a rear-view mirror in vehicles and why?
8. What is the minimum distance needed to hear an echo?
9. State Fleming's Left-Hand Rule (what it applies to).
10. What is 1 horsepower in watts?
11. What radiation has the highest penetrating power — alpha, beta, or gamma?
12. What is the source of the Sun's energy?
13. Who discovered X-rays and in what year?
14. Name three applications of total internal reflection.
15. The sky is blue due to which type of scattering?
16. A transformer works on which principle and which type of current?
17. Which scientist first explained the photoelectric effect?
18. Latent heat of vapourisation of water is approximately how much?
19. What is the commercial unit of electrical energy?
20. Why do stars twinkle but planets do not?
21. Which law of thermodynamics says entropy always increases?
22. Name the person who discovered the neutron.
23. What does SONAR stand for and what does it use?
24. Refractive index of diamond is approximately how much?
25. Indian household electricity supply is at what voltage and frequency?

Answers: 1. Pascal (Pa) 2. Michael Faraday, 1831 3. 11.2 km/s 4. Higher pressure raises the boiling point of water to ~120 °C; food cooks faster 5. Concave (diverging) lens 6. 331 m/s 7. Convex mirror — because it always gives a virtual, erect, diminished image with a wider field of view 8. 17 metres 9. Applies to an electric motor — Index = Field direction, Middle = Current direction, Thumb = Thrust/motion direction 10. 746 watts 11. Gamma radiation 12. Nuclear fusion (hydrogen nuclei fuse to form helium) 13. Wilhelm Röntgen, 1895 14. Optical fibres, diamonds (sparkle), mirages in desert, periscope 15. Rayleigh scattering 16. Mutual induction; AC (Alternating Current) only 17. Albert Einstein (Nobel Prize 1921) 18. 2260 J/g (at 100 °C) 19. Kilowatt-hour (kWh) — also called 1 unit 20. Stars are point sources — atmospheric turbulence flickers their single ray. Planets are extended discs — their multiple rays average out 21. Second Law of Thermodynamics 22. James Chadwick (1932) 23. Sound Navigation and Ranging; uses ultrasonic waves 24. 2.42 25. 230 V, 50 Hz AC

APPENDIX B — Trap Recognition Card

Trap	Common wrong answer	Correct answer
First Nobel Prize in Physics	Einstein, Curie	Röntgen (1901, for X-rays)
Nobel winner for photoelectric effect	J.J. Thomson	Albert Einstein (1921)
Raman Effect Nobel	S.N. Bose	C.V. Raman (1930)
Radio invention — Indian angle	Marconi alone	J.C. Bose also demonstrated millimetre waves in 1895
Sun's energy	Nuclear fission	Nuclear fusion
Most penetrating radiation	Beta	Gamma
What stops alpha radiation	Lead	Paper or a few cm of air
Rear-view mirror type	Concave	Convex (wider field)
Myopia correction	Convex	Concave lens
Speed of sound vs speed of light	Similar	Light is ~880,000× faster
Echo minimum distance	34 m	17 m (half the round trip)
Horsepower in watts	750 W	746 W
Ice floats because...	It is lighter	Anomalous expansion — water expands on freezing
Geostationary satellite height	3600 km	~36,000 km
Transformer works on DC?	Yes	No — only AC creates changing flux